

Complex Analysis Homework #5

due 2026 APR 20, 10:00 a.m.

1 Assignments

All problems are in the handout.

- I. Exercise 3.39;
- II. Exercise 3.41;
- III. Exercise 3.42;
- IV. Exercise 3.51;
- V. Exercise 3.52;
- VI. Exercise 3.58;
- VII. Exercise 3.59;
- VIII. Exercise 3.63.

Each problem weighs 10 points, while Problems III and VII weigh 20 points. The full mark is 100 while the highest possible score is 110; see Section 3.

Caution: please make sure that your handwriting is recognizable, otherwise you only get partial credit for the recognizable part.

重要提醒: 证明时必须说明所用的定理定义等依据, 并且步骤详细, 否则不得分!

2 Tips

The four steps of math learning are:

- L1. understand the ideas, e.g. why the theorems hold,
- L2. memorize the packaging, i.e. the summary handout,
- L3. apply the math to solve problems,
- L4. integrate new learning into your knowledge system.

Doing homework is in the early stage of L3, so it is often time-efficient to do step L2 before attempting the homework problems.

Ask yourself two questions after finishing a problem:

- “What previous knowledge have I used?”
- “What is the connection between the previous knowledge to the new concepts and theorems?”

3 Extra credits

Additional 10% credits will be given to you if you typeset your solutions in $\text{\LaTeX}$. You can also get partial extra credit for typesetting solutions of *some* problems.